

Week -1

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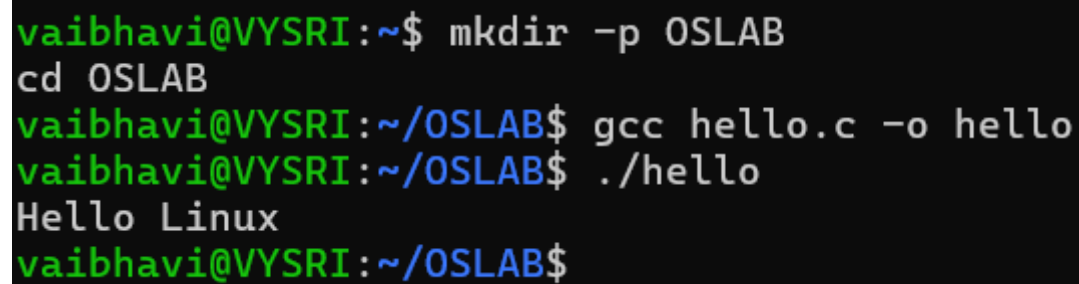
Roll No: 2520030398

1.Hello.c

```
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

OUTPUT:

A terminal window with a black background and green text. The user 'vaibhavi' is at host 'VYSRI' in the home directory '~'. They run 'mkdir -p OSLAB', then 'cd OSLAB'. In the directory '~/OSLAB', they compile 'hello.c' to 'hello' using 'gcc', and then run './hello'. The output is 'Hello Linux'.

```
vaibhavi@VYSRI:~$ mkdir -p OSLAB
cd OSLAB
vaibhavi@VYSRI:~/OSLAB$ gcc hello.c -o hello
vaibhavi@VYSRI:~/OSLAB$ ./hello
Hello Linux
vaibhavi@VYSRI:~/OSLAB$
```

2. Read User Name

```
#include<stdio.h>

int main()
{
    char name[30];
    printf("Enter your name: ");
    scanf("%s",name);
    printf("Welcome %s\n",name);
    return 0;
}
```

Output:

```
vaibhavi@VYSRI:~/OSLAB$ nano readusername.c
vaibhavi@VYSRI:~/OSLAB$ gcc readusername.c -o firstprogram
vaibhavi@VYSRI:~/OSLAB$ ./firstprogram
Enter your name: vaibhavi
Welcome vaibhavi
```

3. Display Process ID

```
#include<stdio.h>

#include<unistd.h>

int main()
{
    printf("Process ID = %d\n",getpid());
    return 0;
}
```

OUTPUT:

```
vaibhavi@VYSRI:~/OSLAB$ nano displayprocessid.c
vaibhavi@VYSRI:~/OSLAB$ gcc displayprocessid.c -o firstprogram
vaibhavi@VYSRI:~/OSLAB$ ./firstprogram
Process ID = 558
vaibhavi@VYSRI:~/OSLAB$
```

4. #include<stdio.h>

```
#include<unistd.h>
```

```
int main()
{
    fork();
    printf("Hello\n");
    return 0;
}
```

OUTPUT:

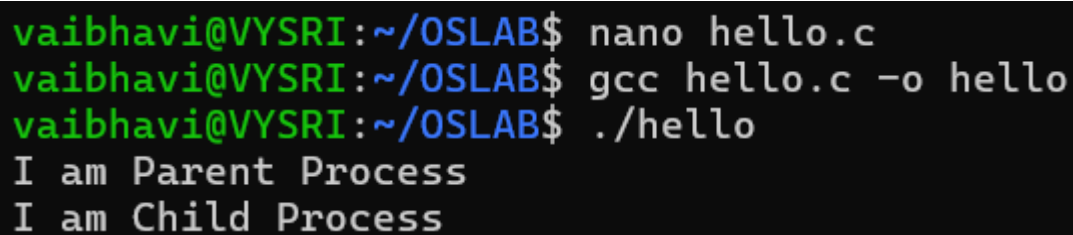
```
vaibhavi@VYSRI:~/OSLAB$ nano fork.c
vaibhavi@VYSRI:~/OSLAB$ gcc fork.c -o firstprogram
vaibhavi@VYSRI:~/OSLAB$ ./firstprogram
Hello
Hello
```

5. Parent and Child

```
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;
    pid=fork();
    if(pid==0)
    {
        printf("I am Child Process\n");
    }
    else
    {
        printf("I am Parent Process\n");
    }
    return 0;
}
```

OUTPUT:

A terminal window with a black background and green text. It shows the user 'vaibhavi' at host 'VYSRI' in the directory '~/OSLAB'. The user runs 'nano hello.c', then 'gcc hello.c -o hello', and finally './hello'. The output of the program is 'I am Parent Process' followed by 'I am Child Process' on the next line.

```
vaibhavi@VYSRI:~/OSLAB$ nano hello.c
vaibhavi@VYSRI:~/OSLAB$ gcc hello.c -o hello
vaibhavi@VYSRI:~/OSLAB$ ./hello
I am Parent Process
I am Child Process
```

6.Display parent and Child PID

```
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;
    pid=fork();
```

```

if(pid==0)
{
    printf("Child PID = %d\n",getpid());
    printf("Parent PID = %d\n",getppid());
}
else
{
    printf("Parent PID = %d\n",getpid());
}
return 0;
}

```

OUTPUT:

```

vaibhavi@VYSRI:~/OSLAB$ nano displayparentandchild.c
vaibhavi@VYSRI:~/OSLAB$ gcc displayparentandchild.c -o firstprogram
vaibhavi@VYSRI:~/OSLAB$ ./firstprogram
Parent PID = 714
Child PID = 715
Parent PID = 300

```

7.Wait()

```

#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>
int main()
{
    pid_t pid;
    pid=fork();

    if(pid==0)
    {
        printf("Child Running\n");
    }
}

```

```

else
{
    wait(NULL);
    printf("Parent Running After Child\n");
}
return 0;
}

```

OUTPUT:

```

vaibhavi@VYSRI:~/OSLAB$ nano hello.c
vaibhavi@VYSRI:~/OSLAB$ gcc hello.c -o hello
vaibhavi@VYSRI:~/OSLAB$ ./hello
Child Running
Parent Running After Child

```

8. Wait two()

```

#include<stdio.h>
#include<unistd.h>
int main()
{
    printf("Before exec()\n");
    execlp("ls","ls",NULL);
    printf("After exec()\n");

    return 0;
}

```

OUTPUT:

```

vaibhavi@VYSRI:~/OSLAB$ nano waittwo.c
vaibhavi@VYSRI:~/OSLAB$ gcc waittwo.c -o waittwo
vaibhavi@VYSRI:~/OSLAB$ ./waittwo
Before exec()
displayparentandchild.c  firstprogram  hello  parentandchild  processid.c  wait.c  waittwo.c
displayprocessid.c      fork.c       hello.c  parentandchild.c  readusername.c  waittwo
vaibhavi@VYSRI:~/OSLAB$

```

9.fork() + exec()

```

#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>
int main()
{
    pid_t pid;
    pid=fork();
    if(pid==0)
    {
        execlp("date","date",NULL);
    }
    else
    {
        wait(NULL);

        printf("Child Completed\n");
    }
    return 0;
}

```

OUTPUT:

```

displayprocessid.c  fork.c  netto.c  parentandch
vaibhavi@VYSRI:~/OSLAB$ nano exec.c
vaibhavi@VYSRI:~/OSLAB$ gcc exec.c -o firstprogram
vaibhavi@VYSRI:~/OSLAB$ ./firstprogram
Wed Aug  5 15:23:26 UTC 2026
Child Completed
vaibhavi@VYSRI:~/OSLAB$

```